

KNN, Metrics, and Regression from Scratch

[FILL IN: Your Name]

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MLE-AI-201 Machine Learning, Cohort II
AI Academy, National AI Center

Abstract—[FILL IN: 3–5 sentences. State what you implemented (KNN, four metrics, linear and logistic regression), the datasets used, your headline numbers (e.g. best k and test F_1 ; linear-regression test R^2 ; logistic-regression test accuracy), and whether your from-scratch code matched scikit-learn.]

Index Terms— k -nearest neighbors, classification metrics, linear regression, logistic regression, gradient descent, regularization.

[FILL IN: replace this box with
`\includegraphics{knn_sweep.pdf}`]

Fig. 1: Validation F_1 as a function of k .

I INTRODUCTION

[FILL IN: One short paragraph. What is the goal of this assignment and what will the reader find in this report? Mention that every algorithm is implemented in NumPy and validated against scikit-learn.]

II KNN AND METRICS

A KNN implementation

[FILL IN: Describe how you vectorized the Euclidean distance (no Python loop over training points). One or two sentences plus the key equation if helpful.]

The pairwise squared distance can be expanded as

$$\|\mathbf{x} - \mathbf{z}\|_2^2 = \|\mathbf{x}\|_2^2 - 2\mathbf{x}^\top \mathbf{z} + \|\mathbf{z}\|_2^2. \quad (1)$$

B Metrics implementation

[FILL IN: Briefly state how you computed accuracy, precision, recall, and F_1 from the confusion-matrix counts, and how you handled the zero-denominator edge case.]

C Results

[FILL IN: Report the metric-vs- k sweep and the best k by F_1 . Reference the table and figure below. Confirm the match with scikit-learn.]

Table 1: Test-set metrics for the best k (breast cancer).

Model	Acc	Prec	Rec	F_1
Your KNN ($k=[..]$)	[..]	[..]	[..]	[..]
sklearn KNN	[..]	[..]	[..]	[..]

III LINEAR AND LOGISTIC REGRESSION

A Linear regression

[FILL IN: Describe your batch gradient-descent update, learning rate, and number of epochs. Report test-set MSE and R^2 on the diabetes dataset and compare with scikit-learn.]

The mean-squared-error objective and its gradient are

$$J(\mathbf{w}, b) = \frac{1}{N} \|\mathbf{X}\mathbf{w} + b - \mathbf{y}\|_2^2, \quad \nabla_{\mathbf{w}} J = \frac{2}{N} \mathbf{X}^\top (\mathbf{X}\mathbf{w} + b - \mathbf{y}). \quad (2)$$

B Logistic regression

[FILL IN: Describe your cross-entropy gradient descent on the breast cancer dataset. Report test-set accuracy, precision, recall, and F_1 (reuse your Part 1 metrics) and compare with scikit-learn.]

The sigmoid and cross-entropy gradient are

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad \nabla_{\mathbf{w}} J = \frac{1}{N} \mathbf{X}^\top (\hat{\mathbf{p}} - \mathbf{y}). \quad (3)$$

C Training curves

[FILL IN: Show and briefly discuss the loss-vs-epoch curves for both models. A decreasing curve confirms the learning rate is sensible.]

IV BONUS: REGULARIZATION (OPTIONAL)

[FILL IN, only if attempted. State clearly which bonus(es) you did. For L1: show that a large λ drives some weights toward zero. For L2: show that weights shrink smoothly. Report how the weight norm and test performance change with λ .]

V CONCLUSION

[FILL IN: 2–3 sentences. What worked, what surprised you, and where your from-scratch results differed from scikit-learn (and why).]

AI-TOOL USAGE DECLARATION

[FILL IN: Name any AI tool used, what you used it for, which parts it affected, and how you checked the output. Write "No AI tools used." if that is the case. This declaration is required by the syllabus.]

REFERENCES

- [1] G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, *An Introduction to Statistical Learning with Applications in Python*. Springer, 2023.
- [2] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. Springer, 2009.
- [3] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *JMLR*, vol. 12, pp. 2825–2830, 2011.